

Week 6 Final Report - Baseline Model Evaluation

Student: Shari Oliver

Project: Carisurg AI Triage Prediction

Week: 6 – Baseline Models and Clinical Evaluation

1. Objective

The objective of this phase was to establish a clinically defensible baseline for predicting Emergency Severity Index (ESI) triage levels using machine learning. Following approval of the Week 5 feasibility assessment, two interpretable models were developed:

- Logistic Regression
- Decision Tree Classifier

Both models were compared against a stratified random baseline to determine whether meaningful predictive performance could be achieved before considering more complex methods.

2. Dataset and Experimental Setup

The cleaned Mercer Emergency Department triage dataset prepared during Week 5 was used for modelling.

An 80/20 stratified train-test split was performed to preserve the distribution of ESI classes in both datasets.

Random Seed: 42

The Decision Tree depth was constrained to reduce overfitting and preserve interpretability for clinical stakeholders.

3. Class Distribution

The dataset is highly imbalanced.

ESI Level	Clinical Meaning	Relative Frequency
ESI 1	Immediate life threat	Extremely rare
ESI 2	Emergent	Common
ESI 3	Urgent	Most common
ESI 4	Less urgent	Moderate
ESI 5	Non-urgent	Rare

As shown in the distribution plot, ESI Levels 2 and 3 account for the overwhelming majority of patients, while ESI Level 1 cases represent only a very small proportion of presentations.

This imbalance makes overall accuracy a potentially misleading evaluation metric.

4. Model Performance

Model	Accuracy	Macro F1	Weighted F1	ESI 1 Recall
Stratified Random Baseline	0.38	0.20	0.38	0.00
Logistic Regression	0.59	0.43	0.62	0.63

Decision Tree	0.25	0.17	0.29	0.56
---------------	------	------	------	------

Logistic Regression

Logistic Regression substantially outperformed the random baseline across every metric.

The model achieved:

- 59% overall accuracy
- Weighted F1 score of 0.62
- ESI Level 1 recall of 62.5%

The confusion matrix shows that predictions generally remain close to the correct severity level, with most errors occurring between neighbouring classes (for example ESI 2 and ESI 3).

This behaviour is clinically more acceptable than large jumps between severity levels.

Decision Tree

The Decision Tree performed poorly despite capturing some ESI Level 1 cases.

The model achieved only:

- 25% accuracy
- 0.17 macro F1 score
- 0.29 weighted F1 score

The confusion matrix shows severe class collapse, with the model heavily over-predicting ESI Level 5 while almost never predicting ESI Level 4 correctly.

This suggests the model failed to generalise and was highly sensitive to the underlying class imbalance.

As a result, the Decision Tree was not considered suitable for deployment as a baseline model.

5. Primary Metric Selection

The primary evaluation metric selected for this project was **recall for ESI Level 1 patients**.

In an Emergency Department setting, the most serious error is failing to identify a critically ill patient who requires immediate intervention.

A patient experiencing cardiac arrest, major trauma, respiratory failure, or septic shock who is incorrectly classified into a lower urgency category may experience delays in treatment that result in significant morbidity or mortality.

While false positives may increase clinician workload, false negatives in ESI Level 1 cases represent a direct patient safety risk.

For this reason, detecting as many true ESI Level 1 patients as possible was considered more important than maximising overall accuracy.

6. Macro F1 versus Weighted F1

Two versions of F1 score were reported:

Macro F1

- Treats every ESI class equally.
- Rare classes contribute as much to the score as common classes.

Weighted F1

- Weights classes according to their frequency in the dataset.
- Dominated by performance on ESI Levels 2 and 3.

The large difference between Logistic Regression's Macro F1 (0.43) and Weighted F1 (0.62) indicates that the model performs substantially better on common classes than on rare classes.

This behaviour is expected given the severe imbalance in the dataset.

7. Failure Modes

The failure mode of greatest clinical concern is the under-triage of ESI Level 1 patients.

The Logistic Regression confusion matrix shows that six ESI Level 1 patients were incorrectly assigned lower acuity categories:

- Five patients were predicted as ESI 2.
- One patient was predicted as ESI 5.

Although an ESI 2 prediction would still likely receive rapid clinical review, the ESI 5 prediction represents a potentially dangerous miss.

This failure mode would require further investigation before deployment.

Potential mitigation strategies include:

- Class weighting during training
- Oversampling rare classes
- Cost-sensitive learning approaches
- Additional physiological predictors

8. Conclusion

The project successfully established a defensible machine learning baseline for Emergency Department triage prediction.

Logistic Regression demonstrated meaningful predictive ability and substantially outperformed both the stratified random baseline and the Decision Tree classifier.

Most importantly, the model correctly identified approximately **63% of ESI Level 1 patients**, demonstrating that even a simple interpretable model can capture clinically relevant patterns within the dataset.

The Logistic Regression model will therefore be carried forward as the benchmark for future development and comparison against more sophisticated approaches.